

Presentation 1

- Link to presentation:
https://github.com/arvindkrishna87/STAT390_Winter2026/blob/main/Team2/Presentations/Slides/Presentation1_Team2.pdf
- Link to code:
https://github.com/arvindkrishna87/STAT390_Winter2026/tree/main/Team2/Code

What did each of the team members do?

Adithi & Jason:

- For presentation 1, we went through the Fall 2025 Final Presentation pdf and the CMIL Classification Problem Statement and Data Winter 2026 presentation pdf linked on Canvas to fully understand what C-MIL is, how it is being diagnosed, and how prior classes have begun creating an ML model to classify different tissue slices as different grades of C-MIL.
- We conducted a literature search on C-MIL and its progressed form of Co-M (conjunctival melanoma) to determine where, biologically, the current ML model implementation may be lacking. Accordingly, we made suggestions regarding stratifying our dataset by age and race.

Roma & Paige:

- We copied to our Team folder and analyzed the most recent version of the code produced by the Fall 2025 project team. Per suggestion in class, we focused on understanding the logic and organization of main.py, and subsequently looked through the referenced python files (ex. utils.py, data_utils.py etc.). We wrote our own line-by-line code notes on the 4 main functions in main.py and also sought to understand the different functions it referenced, to understand the structure, logic, and purpose of the codebase.
 - We watched the 2025 code organization video to help guide our understanding of the code flow and logic.
- We modified all the sbatch files to update the file paths and successfully ran the program. We waited for the code to run over several hours (about 14 hours), then located and identified the different outputs.
- Per Professor Krish's request, we investigated why the values in the "mean/median patches per stain per split" tables were so low and if there was an error in the computation or the annotation.

- We focused on studying how the splitting and stratification logic works in preparation for a potential future task of editing the stratification methods based on our consultants' suggestions.

Charlene:

- I went through the "Synergistic H&E and IHC image analysis by AI predicts cancer biomarkers and survival outcomes in colorectal and breast cancer". I focused on understanding model architecture, experimental setup, training and evaluation strategy, and statistical methodology.
- I considered a few critiques on the model's assumptions and design choices, identifying potential methodological limitations related to reproducibility, generalizability, data sampling, and labeling strategies. Based on this analysis, I proposed targeted critiques and corresponding improvement suggestions that could strengthen model robustness and clinical applicability.

How does it help the project?

Adithi & Jason:

- Bringing in a new perspective, one from the medical/biological side, is very helpful in identifying limitations in the current models and suggesting new solutions that can improve the accuracy of said models. By starting off broadly, with research into the demographics of our dataset, we can first tackle more biologically simple reasons for impacted accuracy all while setting a foundation to build up to more complex concepts in the following weeks.

Roma & Paige:

- By understanding the entire code, we will be able to make effective and accurate changes in the future to improve the model. There are a lot of code, files, and unfamiliar organization, so this was a very necessary and helpful process so we will be able to more effectively navigate the files in the coming weeks.
- By identifying the different types of print-outs and result files, we will be better equipped to point our consultants to the relevant outputs and analyze them ourselves when we implement our suggestions and run our own experiments.
- The corrected annotation suggestion for `print_data_summary()` will just help make sure the interpretability of the model data is clear.

Charlene:

- Understanding the methodology and the setup of our model is important to understand the next steps. By evaluating the current model, I noted certain areas where it could be probable issues which may not be immediately apparent from the performance metrics alone, such as bias from retrospective data, tile sampling limitations, and partial biomarker labeling.
- These considerations will assist us what we want to improve on in the model and provide a framework for evaluating the actual model performance of genuine biological learning or dataset-specific patterns.

Issues faced that haven't been resolved:

Adithi & Jason:

- We do not have access to the full dataset yet. We have discussed this with Krish and will have access before the next presentation. This will allow us to make more targeted follow-up recommendations, again from the biological perspective.

Roma & Paige:

- We had difficulties logging into quest Saturday, 1/31 and want to investigate if there is an access issue.

Charlene:

- There is limited access to the current project dataset restricts the ability to directly assess whether the identified methodological concerns apply to the C-MIL task.

Proposed next steps:

Adithi & Jason:

- Based on feedback we received from Krish in class, we will dive deeper into the dataset to determine if it is possible to consider location of melanocyte cell-bulks within (and beyond) the intraepithelial region as a way to better determine C-MIL grade.

Roma & Paige:

- Work with consultants to make model adjustments based on their suggestions and objectives for next week.

Charlene:

- Next steps is that I will evaluate the model to identify similar risks. Then I would use this information to examine the performance trends across folds, subgroups and tissue characteristics to assess the model's robustness and generalizability.

References

- Fall 2025 Final Presentation:
- Fall 2025 Code Explanation:
<https://nuwildcat-my.sharepoint.com/my?id=%2Fpersonal%2Fakl0407%5Fads%5Fnorthwestern%5Fedu%2FDocuments%2FBack%20up%2F2025%2D26%2FWinter%202026%2FSTAT390%2FStudent%20presentations&ga=1>
- CMIL Classification Problem Statement and Data Winter 2026 presentation
- WHO Paper
- Butt et al source
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC12448771/>
- “Synergistic H&E and IHC image analysis by AI predicts cancer biomarkers and survival outcomes in colorectal and breast cancer”
- Nita Mulliqi, Kimmo Kartasalo, Henrik Olsson, Xiaoyi Ji, Lars Egevad, Martin Eklund, Pekka Ruusuvuori, OpenPhi: an interface to access Philips iSyntax whole slide images for computational pathology, *Bioinformatics*, Volume 37, Issue 21, November 2021, Pages 3995–3997, <https://doi.org/10.1093/bioinformatics/btab578>